



UNIVERSITY OF
CAMBRIDGE

sth like Mechanistic Interpretability of Large Language Models

Elisabeta-Iulia (Julia) Dima



King's College

This dissertation is submitted on June, 2026 for the degree of Master of Philosophy

Abstract

My abstract ...

Declaration

This dissertation is the result of my own work and includes nothing which is the outcome of work done in collaboration except as declared in the Preface and specified in the text. It is not substantially the same as any that I have submitted, or am concurrently submitting, for a degree or diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the Preface and specified in the text. I further state that no substantial part of my dissertation has already been submitted, or is being concurrently submitted, for any such degree, diploma or other qualification at the University of Cambridge or any other University or similar institution except as declared in the Preface and specified in the text. This dissertation does not exceed the prescribed limit of 60 000 words.

Elisabeta-Iulia (Julia) Dima

June, 2026

Acknowledgements

My acknowledgements ...

Contents

1	Introduction	11
1.1	Mechanistic Interpretability	11
2	Related Work	13
2.1	Arithmetic algorithms in transformers	13
	Bibliography	15
	Glossary	17

Chapter 1

Introduction

Understanding what neural networks actually *compute*, rather than what they *output*, is the central problem of mechanistic interpretability. A model that correctly solves arithmetic problems or chains reasoning steps coherently need not be implementing anything resembling the algorithm a human would use. Whether it does, where in the network that computation lives, and whether it can be manipulated predictably are empirical questions that evaluation benchmarks cannot answer.

1.1 Mechanistic Interpretability

Anthropic’s *On the Biology of a Large Language Model* [1] proposes a reproducible methodology for approaching such questions on a large language model (LLM). Combining sparse autoencoders (SAEs) with cross-layer transcoders (CLTs) and backward-pass attribution, it constructs dependency graphs that trace information flow from input tokens through intermediate features to output logits. The framework is the starting point of mechanistic interpretability work done in this thesis, focused on how mathematical primitives are implemented by LLMs, and how these implementations can be improved through insights from mechanistic interpretability.

Prior mechanistic work has established that small transformers trained on arithmetic tasks converge on specific and inspectable algorithms rather than fitting statistical regularities [3, 6, 4, 5, 2]. Whether analogous structure arises in a large instruction-tuned model, trained without an explicit arithmetic curriculum on substantially more diverse data, is considerably less clear and is a central concern of this work. Furthermore, since mechanistic interpretability offers relevant insights into how these models perform algorithmic tasks, we can make use of this knowledge to improve model performance on algorithmic tasks, which is the focus of the second part of this thesis.

We take the open access model `Qwen3` as our subject model and integer addition as our primary task. Addition is a reasonable prototype due to having unambiguous

ground truth and being structured enough that prior work has identified non-trivial internal mechanisms [3, 4], and performance degrades in interpretable ways as digit magnitude increases. We initially characterise the model’s behaviour on addition using the existing attribution graph methodology given by Lindsey et al. [1]. Beyond reproducing the attribution graphs, we perform intervention studies such as constrained patching to validate hypotheses regarding the underlying computation. We also use linear probes to localise where arithmetic-relevant information is encoded across layers.

By drawing insights from our mechanistic analysis, we then proceed to improve model performance on mathematical primitives through modifying specific components of the model.

The rest of this dissertation is structured as follows. Chapter 2 covers the relevant background. Chapter ?? describes the experimental setup and behavioural baselines. Chapters ?? through ?? present the mechanistic experiments. Chapter ?? synthesises the findings.

Chapter 2

Related Work

2.1 Arithmetic algorithms in transformers

A productive line of mechanistic interpretability research has asked not merely whether transformers can perform arithmetic, but *how* they do so: what internal representations they form and what computational steps those representations support.

Nanda et al. [3] studied a one-layer transformer trained on modular addition and found that grokking, the phenomenon whereby generalisation appears suddenly after extended training on memorised data, corresponds to the emergence of a Fourier-based algorithm. The trained model encodes operands as superpositions of trigonometric features and composes them across layers to compute the result; crucially, the emergence of this structure can be tracked through interpretable progress measures even before the accuracy curve breaks upward.

Zhong et al. [6] revisited modular addition and showed that the same task admits two qualitatively distinct algorithmic solutions: a Fourier-based strategy they term the “clock” algorithm, and a lookup-table strategy they term the “pizza” algorithm. Which solution emerges depends on training configuration, demonstrating that identical supervision can give rise to structurally different circuits and that mechanistic diversity is not merely an artefact of random initialisation.

For standard integer addition, Quirke and Barez [4] identified a carry-propagation circuit distributed across attention and MLP layers, bearing a structural resemblance to the long-addition algorithm familiar from elementary arithmetic. Quirke et al. [5] extended this account to subtraction, characterising how the same underlying circuit adapts to accommodate sign.

Musat [2] approached the problem from the perspective of training dynamics, tracing how the internal representations of operands and results cluster and align over the course of training on modular addition. Their analysis shows that organised algorithmic structure is not installed gradually but emerges through identifiable phase transitions, reinforcing

the view that grokking marks a qualitative change in the model’s internal organisation rather than a smooth accumulation of generalising evidence.

Taken together, these results establish a strong prior that transformers trained on arithmetic tasks do not merely interpolate over training examples but converge on discrete, inspectable computational procedures. All of the models studied, however, are small transformers trained from scratch on a single arithmetic task. The present work asks whether comparable structure is recoverable in **Qwen3**, a multi-billion-parameter instruction-tuned model trained on a broad natural-language corpus with no explicit arithmetic curriculum.

Bibliography

- [1] Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Shan Carter, Tom Henighan, Adam Stevenson, Tom Conerly, Danny Hernandez, Adam Jermyn, Callum McDougall, et al. On the biology of a large language model. 2025. URL <https://transformer-circuits.pub/2025/biology-of-a-lm/index.html>. Anthropic Transformer Circuits Thread.
- [2] Tiberiu Musat. Clustering and alignment: Understanding the training dynamics in modular addition, 2024.
- [3] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2301.05217>.
- [4] Philip Quirke and Fazl Barez. Understanding addition in transformers. In *International Conference on Learning Representations*, 2024. URL <https://arxiv.org/abs/2310.13121>.
- [5] Philip Quirke, Clement Neo, and Fazl Barez. Understanding addition and subtraction in transformers, 2025.
- [6] Ziqian Zhong, Ziming Liu, Max Tegmark, and Jacob Andreas. The clock and the pizza: Two stories in mechanistic explanation of neural networks. In *Advances in Neural Information Processing Systems*, 2023. doi: 10.48550/arXiv.2306.17844.

Glossary

Attribution graph A directed acyclic graph whose nodes represent model components (token embeddings, SAE or CLT features, attention heads, output logits) and whose edges represent gradient-weighted information flow between them. Introduced as a methodology by Lindsey et al. [1].

Circuit A subgraph of the model’s computational graph comprising a small set of components (attention heads, MLP layers, or individual features) that together implement a specific behaviour. The term is used in mechanistic interpretability to denote minimal, interpretable sub-computations.

CLT — Cross-Layer Transcoder A sparse dictionary-learning module, analogous to a sparse autoencoder, that maps MLP inputs at one layer to residual-stream contributions, potentially spanning across layers.

FFN — Feed-Forward Network Synonym for MLP in the transformer literature.

Grokking The phenomenon whereby a neural network trained on a small dataset first memorises the training examples (near-perfect training accuracy, poor generalisation) and then, after many additional gradient steps, abruptly transitions to a generalising solution. Mechanistically characterised for modular arithmetic by Nanda et al. [3].

LLM — Large Language Model A neural language model with a parameter count large enough that emergent capabilities arise from scale alone, typically trained on broad natural-language corpora via next-token prediction.

MLP — Multilayer Perceptron In the transformer architecture, the feed-forward sub-layer of each block, consisting of two learned linear projections with a pointwise nonlinear activation in between. Despite the name, transformer MLPs typically have only a single hidden layer.

RMSNorm — Root Mean Square Layer Normalisation A variant of layer normalisation that rescales activations by their root mean square without subtracting the mean. Used in Qwen and other modern architectures in place of standard LayerNorm.

SAE — Sparse Autoencoder A dictionary-learning module trained to decompose neural network activations into a sparse linear combination of learned feature directions. Used in mechanistic interpretability to identify human-interpretable features inside a model.